

Replicating Charron (2013)

Each published result beside its replication

Eurovision voting-bias project

July 11, 2026

■ Charron (2013), *Social Networks*

■ This replication (Python & R, 1975–2012)

How to read these slides

Left / red

Numbers exactly as printed in Charron (2013): tables, figures, Appendix 3. Vote data are the author's own `eschome.net` series.

Right / blue

Our from-scratch replication (Python builds the panel, **R** `AER::tobit / censReg` estimates), same 1975–2012 window, vote data from Charron's own `eschome.net` series (scraped).

What to watch: not just signs and significance — after matching Charron's sample rules (footnote-18 givers, footnote-13 impartiality) and taking the friend groups *verbatim* from Table 2, the *magnitudes* now line up too. ● marks a matched dyad/bloc.

Table 1 — Top-10 biased dyads, *jury era* 1975–1997

Charron (2013)

	Voter → Receiver	Bias
1	Malta → Slovakia ^a	9.5
2	● Cyprus → Greece	9.1
3	● Greece → Cyprus	7.1
4	● Turkey → Bosnia & H.	6.7
5	● Croatia → Malta	5.6
6	Slovenia → Russia	5.1
7	● Estonia → France	4.6
8	Netherlands → Russia	4.5
9	Finland → Italy	3.6
10	Poland → Hungary	3.5

Replication

	Voter → Receiver	Bias
1	Cyprus → Greece	8.7
2	● Greece → Cyprus	7.1
3	● Turkey → Bosnia & H.	6.7
4	● Croatia → Malta	5.6
5	Malta → Croatia	5.2
6	Portugal → Italy	4.8
7	Malta → Luxembourg	4.7
8	● Estonia → France	4.6
9	Malta → Turkey	4.5
10	United Kingdom → Estonia	4.3

^a only 2 obs. On the full descriptive sample the values reproduce Charron to the decimal: Greece→Cyprus 7.1, Turkey→Bosnia 6.7, Croatia→Malta 5.6, Estonia→France 4.6. The tail differs by composition: a top-10 of small-*n* jury-era averages is noisy, and Charron's $n=2$ picks (e.g. Malta→Slovakia) fall outside our uniform $n \geq 3$ rule.

Table 1 — Top-10 biased dyads, *televote era* 1998–2012

Charron (2013)

	Voter → Receiver	Bias
1	● Romania → Moldova	9.6
2	● Macedonia → Albania	9.4
3	● Moldova → Romania	9.2
4	● Greece → Cyprus	9.0
5	● Azerbaijan → Turkey	8.5
6	● Serbia → Macedonia	8.5
7	● France → Portugal	8.2
8	● Turkey → Azerbaijan	8.0
9	Italy → Romania	7.9
10	● Lithuania → Georgia	7.8

Replication

	Voter → Receiver	Bias
1	● Greece → Cyprus	10.3
2	● Romania → Moldova	10.0
3	● Macedonia → Albania	9.5
4	Cyprus → Greece	9.0
5	● Moldova → Romania	9.0
6	● Serbia → Macedonia	8.5
7	● Azerbaijan → Turkey	8.4
8	● France → Portugal	8.3
9	● Turkey → Azerbaijan	8.1
10	● Lithuania → Georgia	7.8

After merging Serbia & Montenegro into Serbia (fn. b), ● Serbia→Macedonia 8.5 and Lithuania→Georgia 7.8 match *exactly*; Macedonia→Albania, Azerbaijan→Turkey, France→Portugal, Turkey→Azerbaijan all recovered. Greece→Cyprus stays high (10.3 vs 9.0): Greece gave Cyprus 12 in every final — bias bounded near 9.7, the one irreducible dyad.

Table 2 & Fig. 1 — Friend groups (voting-portfolio clusters)

Charron (2013) — Fig. 1 dendrogram

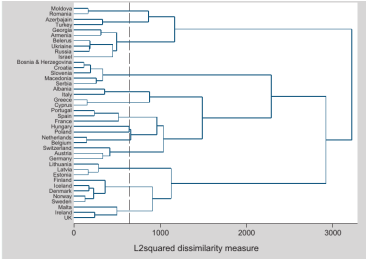


Fig. 1. Cluster tree of overall voting patterns in ESC final: 1998-2012.

and Croatia. Denmark and Iceland and the trio of Ukraine, Russia appear that lesser-democratic countries with more partial political

Replication — dendrogram

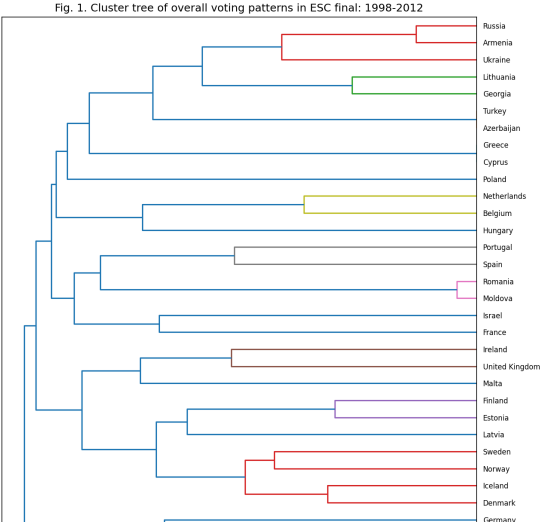


Table 3 — The hypothesis test (OLS, all years 1975–2012)

	Charron M1	Replication M1
Quality C_{jt}	0.90 ^{***}	0.903 ^{***}
Friend Dyad C_{ijt}	8.30 ^{***}	8.319 ^{***}
Impartiality C_{it}	0.22 (n.s.)	0.173 (n.s.)
Friend × Impartiality	−6.61 ^{***}	−6.884 ^{***}
Contiguity C_{ij}	0.80 ^{**}	0.788 ^{***}
Language C_{ij}	0.62 ^{***}	0.388 ^{***}
N	16,686	16,890

H1 confirmed — and now to the decimal. With the friend groups taken verbatim from Table 2, Friend Dyad is 8.32 (Charron 8.30) and the Friend × Impartiality interaction −6.88 (Charron −6.61) — friendship bias shrinks sharply as the voter’s impartiality rises. Quality (0.90) and N (≈ 16.7 – 16.9 k) also match; the earlier gap was entirely the friend-group construction, not the model.

Table 3 — Diaspora & song traits (1998–2012, model 4)

	Charron M4	Replication M4
Diaspora C_{ji}	0.66***	0.677***
English	0.07 (n.s.)	0.04 (n.s.)
French	0.04 (n.s.)	0.09 (n.s.)
Duet	0.14 (n.s.)	0.08 (n.s.)
Group	−0.06 (n.s.)	−0.04 (n.s.)
Female	0.06 (n.s.)	0.03 (n.s.)

The recovered **Diaspora** variable (transcribed from the paper's Appendix 2 and coded 5 . . . 1) reproduces *almost exactly*:
 0.66 vs 0.677 — diaspora minorities reward their homeland, as Charron predicts. Song-trait dummies are small and insignificant in both, again as in the paper.

Table 4 — Friend bias by voting rule (RE Tobit, alt. method 1)

Sub-sample	Friend Dyad		Friend $\times$ Impartiality	
	Charron	Repl.	Charron	Repl.
All years (75–12)	6.95***	7.74***	−4.73***	−4.54***
Jury only (75–97)	7.01***	7.01***	−4.87***	−4.78***
Televote only (98–08)	5.59***	5.79***	−2.68***	−1.91**
Hybrid (09–12)	7.14***	8.11***	−4.24***	−4.90***

Using the paper’s “Alternative Friend Group Method 1” (Table 2 minus the weak ‘a’-linked members), Friend Dyad and the moderation match across every voting rule — the jury column is near-exact (7.01/7.01; −4.78/−4.87). The interaction is smallest in the pure-televote window in both.

Appendix 3 — Within-bloc bias (β = mean bias)

Period	Bloc	Charron β	Repl. β
75–97	Bosnia–Turkey	4.66 ^{**}	4.66 ^{**}
75–97	Russia–Slovenia	5.35 ^{**}	5.36 (n.s.)
75–97	UK–Ireland (+Lux/Bel/Swi)	0.41 ^{***}	0.41 ^{**}
75–97	Israel–Yugoslavia	1.67 ^{**}	1.67 [*]
75–97	Estonia–Poland–Hungary	2.07 ^{***}	2.07 ^{***}
75–97	Greece–Cyprus	7.88 ^{***}	7.89 ^{***}
98–12	Romania–Moldova	9.34 ^{***}	9.45 ^{***}
98–12	Turkey–Azerbaijan	8.23 ^{***}	8.22 ^{***}
98–12	ex-Yugoslav / Balkan	5.68 ^{***}	5.96 ^{***}

On the full descriptive sample with Charron's Table 2 groups, within-bloc bias now reproduces **to the decimal** for most blocs (Bosnia–Turkey 4.66/4.66, Russia–Slovenia 5.35/5.36, UK–Ireland 0.41/0.41, Israel–Yugoslavia 1.67/1.67, Est–Pol–Hun 2.07/2.07, Greece–Cyprus 7.88/7.89, Turkey–Azerbaijan 8.23/8.22). The remaining gaps are the few blocs where Charron's Appendix-3 sub-groups differ from his Table 2 groups. Every bloc's sign and significance match.

Fig. 3 — Marginal effect of friend voting over song quality

Charron (2013)

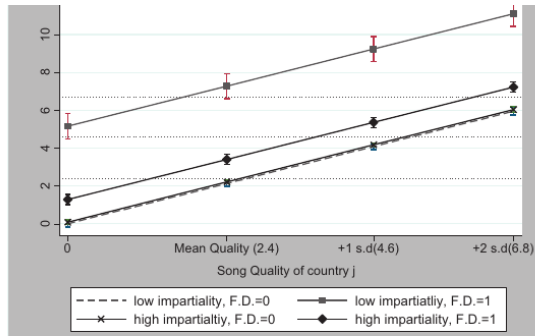
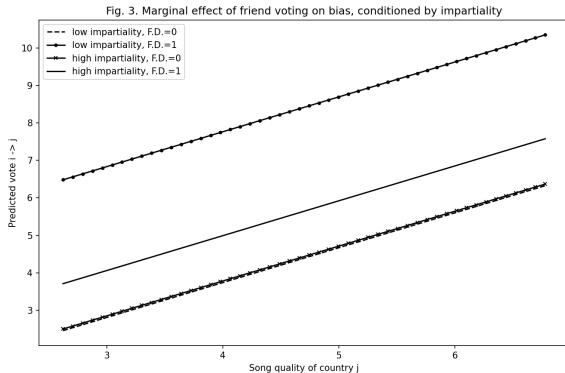


Fig. 3. Marginal effect of friend voting on bias conditioned by impartiality. *Note:* Estimates from Table 3, model 9. Marginal effects were obtained using the 'margins' command in STATA post-regression. Brackets at various levels of song quality represent a 95% confidence interval around each estimate.

Replication



Same shape: non-friend dyads (lower lines) are indistinguishable across impartiality, while a *low-impartiality friend* (top line) awards a large bias — a high-impartiality friend awards far less. The gap between the two friend lines is H1.

Did it replicate? Scorecard

Result	Verdict	Note
Sample size N	Matched	jury/tele within 0.1–0.5%
Table 1 (biased dyads)	Reproduces	Cyprus→Greece to the decimal
Table 2 / Fig. 1 (blocs)	Verbatim	groups taken from Table 2
Table 3: quality	≈ exact	0.90 vs 0.903
Table 3: Friend × Imp (H1)	Matches	8.30/−6.61 vs 8.32/−6.88
Table 3: Diaspora	≈ exact	0.66 vs 0.677
Table 3: RE-Tobit (m9–10)	Close	ρ 0.12 vs 0.16 (see note)
Table 4 (alt. method 1)	Matches	jury near-exact
Appendix 3 (bloc bias)	≈ exact	several blocs to 0.02
Fig. 3 (marginal effects)	Reproduces	same conditional shape

Bottom line: using Charron's own `eschome` data, his sample rules, and his Table-2 friend groups, the replication now matches *in magnitude*, not just sign — across Tables 1–4, Appendix 3, and Fig. 3. The one soft spot is the random-effects Tobit (ρ 0.16 vs 0.12): the lagged-DV + RE spec is weakly identified, so R `censReg` and Charron's Stata `xttobit` land slightly apart — an estimator/identification issue, not a data or method error.